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Alexandre Peres Oliveira da Silva – 20096284

Okeke Munachukwudinamma Praise – 20102055

Spencer Ibhafidon John – 20099070

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Introduction

This report was produced as part of Continuous Assessment 1 for the Advanced Databases module (B9IS125) at Dublin Business School. The aim of this report was to design and analyse relational database systems for a chosen organisation. The organisation chosen for this project is a large public library within Dublin city. For this purpose, an Entity-Relationship Diagram (ERD) has been created to represent the basic elements involved in the functioning of the library. An ERD acts as a model of the database and highlights the relationships among different tables, thereby maintaining consistency within the database. In this report, the group highlights certain important aspects of database design that have been incorporated during the design of the database model.

Organisation Overview

After some deliberation and study of possible organisations, the type of organisation chosen to be the subject of this work is a library. In particular, a public library in a large city, contains a large collection of books and provides lending services to a large number of people. The real world entity which the theoretical library in this work should model is [1]. It was decided that this library, being a public service, does not charge customers money when borrowing a book, nor does it require a membership fee; the only payment required are fines, either for failing to return a book within the appropriate time or for losing a borrowed book. However, it requires customers to be registered as members, as it is necessary to keep records on who is borrowing each book, but it was decided that the library should not keep any more information on possible clients than basic identifying information and a means to get in contact if necessary.

Research into the workings of a library was done throughout this project; at first it was decided that a book's unique identifier should be its ISBN (International Standard Book Number). However, as the database design was developed, it became evident that the ISBN would not be sufficient to identify books within the database. The ISBN is a number unique to a book/edition of a book, yes, but as the database was being developed, not to store various units

of the same book in a single record, but instead each individual physical book, a more precise identifier was required.

After more research, it was decided that the identifying key for each physical book should be its barcode, in a system like what most real libraries use to identify its individual books. The type of barcode chosen for the project was Code 39, a variable length barcode consisting of letters and numbers, since, alongside Codabar, it's a commonly used barcode type in real libraries. At first, it was decided that the ISBN should, instead, be stored as merely one attribute of a book, rather than its identifier, but more research led to the decision of discarding that idea in favour of storing its call number instead, a number that informs the physical location a book should be stored within the library. Call numbers can be based on one of many popular library classification systems, such as the Dewey Decimal System, Universal Decimal Classification and Library of Congress Classification. As the Dewey Decimal System has been noted as commonly used in Ireland, it was decided that the classification system on which the call numbers should be based on.

Books were also given two speciality attributes, which the system should use to record its alternative titles and synopses, blurbs, important reviews, etc. Knowing a book's alternative titles, such as those in other languages and other editions, is helpful to finding alternatives in case a particular book is not in stock, while the blurb, synopses, reviews by relevant individuals or organisations, etc. are crucial for members intending to borrow a book that they are not already familiar with.

In regards to membership, it was decided that there should be different types of membership. For example, students should be prioritized when borrowing books in their topics of study, professors or researchers might need to borrow many books simultaneously, the fines incurred by a child's delayed return of a book should be more lenient and their parent or guardian should bear the fines in their stead. A person may change the type of their membership, but they may only have one type active at a time.

Advantages of DBMS to a Library

The larger an organisation is, the more difficult and impractical it is to keep paper records, doubly so when the organisation needs to keep track of a large number of items, such as the books in a library. Paper records must be manually filled and cautiously stored, and it could take only one wrongly filled or misplaced document to completely lose track of a book. Moreover, paper records are voluminous to store and must be physically consulted, making sharing information among different librarians cumbersome.

Using a digital database, therefore, has a number of advantages to a public library:

1. **Ease to create and maintain records:** Once a digital record is created, it can be easily referenced by accessing the database. That information is preserved and stored in the database, no matter how much time passes between the creation of the record and the time of consulting it. There also is no need to re-enter information that already exists, therefore reducing the chance of inconsistencies arising from wrongly registered information.
2. **Lower space requirements:** Physical records require space; the larger the collection, the more space that paper records require. Digital records, on the other hand, don't occupy physical space; the total space required to store records on the library's entire collection would be, at most, the size of a machine or server where the database is hosted.
3. **Ease to access records:** A digital record can be accessed by any machine with permission, it can even be hosted online. Any information in the database can be easily looked up with a search, instead of requiring a librarian to sift through paper documents, and information can be accessed by multiple machines simultaneously.
4. **More safely stored records:** While a paper document can easily be misplaced or lost, digital databases have many measures to ensure that records are not lost or altered unduly, such as storage redundancy, possible rollbacks and logs.

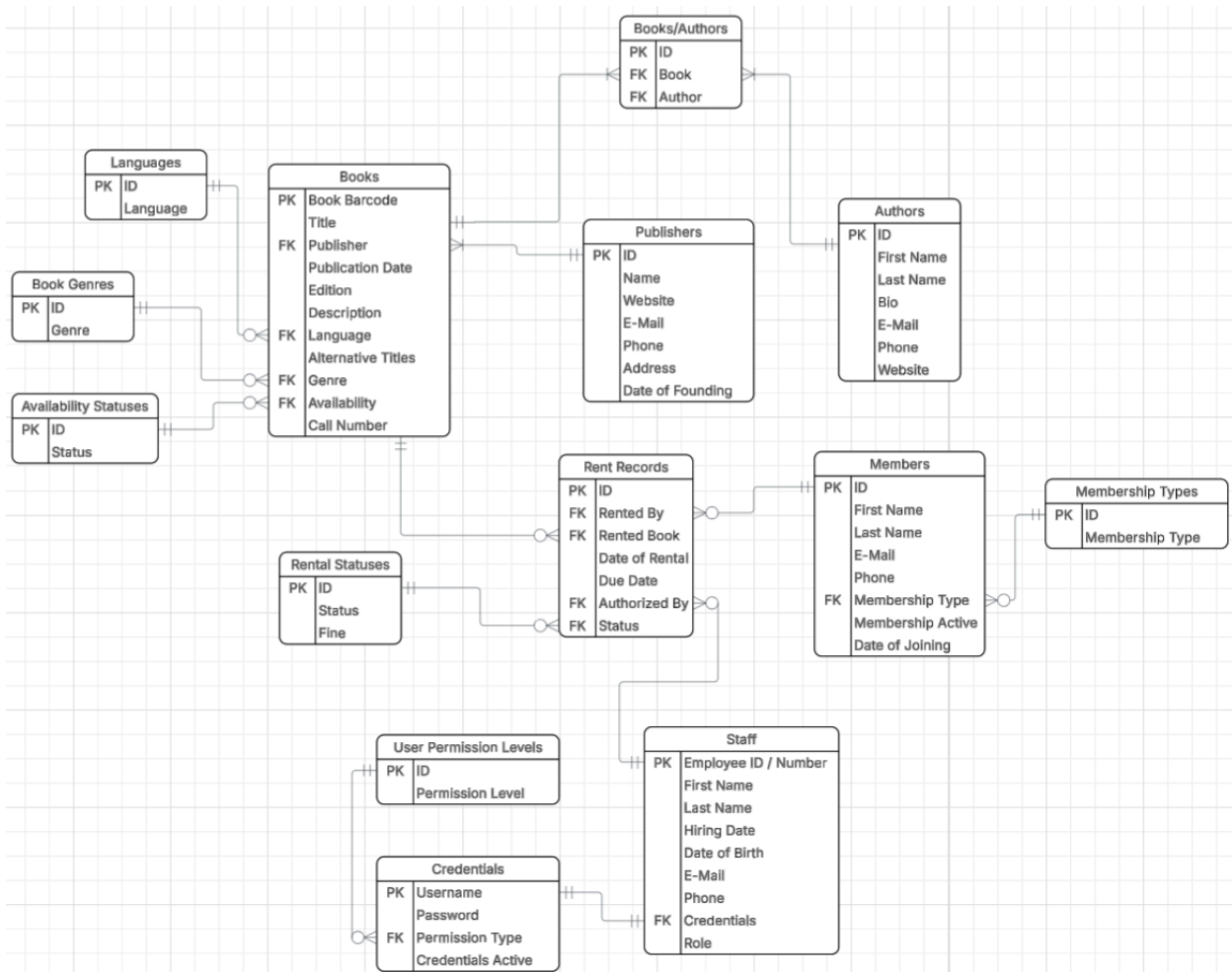
Database Use Cases and Transactions

Operations and transactions that should be supported by the library's database as designed should be:

1. **Add new book:** Register a new physical book to the database, with all the relevant information related to the book, and proper edition, in question.
2. **Add new author:** Register information about a new author to the database.
3. **Add new publisher:** Register information about a new publisher to the database.
4. **Add new member:** Register a new library member to the database.
5. **Add new staff member and create login:** Register a new employee, or staff member, of the library to the database, and, along with it, store the new staff member's system login credentials.
6. **Update author:** Change an author's information, such as phone number, e-mail, website or biography.

7. **Update publisher:** Change a publisher's information, such as phone number, e-mail and website.
8. **Update staff:** Change a staff member's information, such as phone number, e-mail, or role.
9. **Update member:** Change a member's information, such as phone number and e-mail.
10. **Change member's membership type:** Change a member's membership type, such as, a member who is no longer a student must change their membership type to regular adult membership, or a member who has recently enrolled could request to change their membership type to student.
11. **Change member's membership status:** Change a member's membership status between active (has a library card and is permitted to rent books) and inactive (not permitted to rent books, for any reason).
12. **Change book availability status:** Change a book's availability status between available, on hold, or rented.
13. **Change staff member's login permissions:** Change a staff member's credential permissions, that is, what they're authorized to do in the system, in case said staff member is promoted to a position with different permissions.
14. **Remove staff and deactivate login:** Remove the record of a leaving employee, and change their login credentials to inactive.
15. **Rent book:** Create a new rent record, registering the process of a book being rented by a member, the date of the rental and date the book must be returned by.
16. **Change status of book rental:** Change the rent status of a rent record.
17. **Change book's due date:** Change the date the book must be returned by, such as in the case of a renewal.

ERD Diagram



(Diagram created with LucidChart)

Business Rules

Books

1. When a book is registered, it must contain a bar code, title, publication date, edition, language, genre and call number.
2. When a book is registered, it must be associated with at least one author, and one (and only one) publisher.
3. When a book is registered, its availability status is automatically set as "Available".

4. A book's description and alternative titles may be added or edited after creation, and its availability status may be changed after creation.
5. Other than description, alternative titles and availability status, the book's attributes may not be changed outside of fixing typing mistakes.
6. A book's publication date must be a valid date and must be before the date when the book is registered

Authors

1. When an author is registered, they're given an automatically generated ID by the library's system.
2. When an author record is created, it must contain a first name, last name, E-mail and phone number.
3. An author's phone and E-mail must be valid, and not the same as any other author in the system.
4. An author's bio and website may be added or edited after creation.
5. An author's contact information (E-mail and phone number) may be edited after creation, but may never be left blank.
6. An author must be registered to the system when, and only when, the first book by that author is added to the system. An author may not be registered to the system if they are not associated with any books in the system.

Books/Authors

1. Books/authors is the junction table between books and authors. A record in this table is created any time a new book is added to the system, and thus, the connection between the book and its author or authors must be registered in the books/authors table.
2. When a new book/author junction is created to the system, it's given an automatically generated ID by the library's system.
3. A new book/author junction created must represent the relationship between one, and only one, author and one, and only one, book.
4. A book may have multiple authors, and an author may have multiple books. Each one of these relationships must be represented by a separate entry on the books/authors table.

Publishers

1. When a publisher is registered, it's given an automatically generated ID by the library's system.
2. When a publisher is registered, it must contain a name, website, E-mail, phone number, address and date of founding.
3. A publisher's date of founding must be a valid date and must be before the record's creation.
4. A publisher's phone number, E-mail, address and website must be valid, and not the same as any other publisher in the system.
5. A publisher's E-mail, phone number, address and website may be edited after creation, but may never be left blank.
6. A publisher must be registered to the system when, and only when, the first book by that publisher is added to the system. A publisher may not be registered to the system if they are not associated with any books in the system.

Languages

1. When a new language is registered to the system, it's given an automatically generated ID by the library's system.
2. When a new language is registered to the system, its language field must be filled and it must not be the same as any other language in the system.

Book Genres

1. When a new book genre is registered to the system, it's given an automatically generated ID by the library's system.
2. When a new book genre is registered to the system, its genre field must be filled and it must not be the same as any other genre in the system.

Availability Status

1. When a new availability status is registered to the system, it's given an automatically generated ID by the library's system.
2. A book's availability status may only be "Available", "Rented", "On Hold" or "Unavailable for Renting".

Members

1. When a new member is registered to the system, they're given an automatically generated ID by the library's system.
2. When a new member is registered, the record must contain the member's first name, last name, E-mail, phone number and have a membership type.
3. A member's phone and E-mail must be valid, and not the same as any other member in the system.
4. When a new member is registered, their date of joining is set to the day of, and their membership active status is set to "True", that is, active.
5. A member's membership active status may only be "True", meaning active, or "False", meaning inactive.
6. A member's contact information (E-mail and phone number) may be edited after creation, but may never be left blank.
7. A member must always be associated with one, and only one, membership type, however, that type can be changed after creation.

Membership Types

1. When a new membership type is registered to the system, it's given an automatically generated ID by the library's system.
2. A member's membership type may only be "Child", "Young Adult", "Adult", "Student", "Professor", "Researcher" or "Visitor/Tourist".

Staff

1. When a new staff member is registered to the system, they're given an automatically generated ID by the library's system.
2. When a new staff member is registered, their record must contain the staff member's first name, last name, E-mail, phone number, date of birth and role.
3. A staff member's phone and E-mail must be valid, and not the same as any other staff member in the system.
4. When a new member is registered, their date of joining is set to the day of.

5. A staff member's contact information (E-mail and phone number) and role may be edited after creation, but may never be left blank.
6. When a new staff member is registered to the system, their unique credentials must be created and registered to the credentials table. The newly created credentials must then be connected to the new staff member's entry.

Credentials

1. An username must be created for a new staff member at the moment of registering them into the system. That username must be unique within the system, as it is the primary key and identifier of that staff member's credentials.
2. When a new staff member's credentials are registered on the system, they must also create a password. An username must not be registered to the database without an associated password. A new staff member's credentials must also be associated with a permission level at the time of registration.
3. A staff member's password may be edited after creation, but may never be left blank.
4. Passwords must not be registered to the system in plaintext, but instead, its hash value should be calculated with the SHA-256 algorithm, and that hash value is registered instead.
5. When a staff member's credentials is registered to the system, the credentials's active status is set to "True", that is, active.
6. A staff member's credentials' active status may only be "True", meaning active, or "False", meaning inactive.
7. A Staff member's credentials' permission level may be edited after creation, but credentials must always be associated with one, and only one, permission level at any point in time.
8. When a Staff member is suspended, on leave, vacation or let go, their Credentials's Active status should be changed to "False", that is, inactive. When a Staff member returns from suspension, leave or vacation, their Credentials's Active status should be changed to "True", that is, active.

User Permission Levels

1. When a new permission level is registered to the system, it's given an automatically generated ID by the library's system.

2. When a new permission level is registered to the system, its permission level field must be filled and it must not be the same as any other permission level in the system.

Rent Records

1. When a rent record is created, it is given an automatically generated ID by the library's system.
2. When a rent record is created, it must reference one, and only one, book, one, and only one, member and one, and only one, staff member. All this information must be present at the moment of creation.
3. When a rent record is created, its status must be either "On Hold" or "On Time".
4. When a rent record is created, its date of rental is set as the day of.
5. A rent record's due date is calculated on creation, using factors such as, the book's genre, the member's membership type, and how many similar books are on hold.
6. A rent record's rented book, member who rented it, and staff member who authorized it may not be changed outside of fixing typing mistakes.
7. The due date of a rent record may only be changed when a renewal is performed. If the book is returned, a new rent record must be created next time it is rented, even if it's rented again by the same person. The due date of a rent record must never change to a date sooner than its current due date.
8. A rent record's status may only change in the following cases:
 - 8.1. A rent record whose status is "On Hold" can only be updated to "On Time" (the book was rented) or "Returned" (the customer no longer wishes to rent the book).
 - 8.2. A rent record whose status is "On Time" can only be changed to "Overdue" (the book has not yet been returned and it is past the due date), "Returned" (the book was returned) or "Book Lost" (the customer has reported that they have lost the book and will be penalized as such). If a book is renewed while "On Time", its status does not change, only the due date.
 - 8.3. A rent record whose status is "Overdue" can only be changed to "On Time" (in case the book is renewed), "Returned" (the book was returned) or "Book Lost" (the customer has reported that they have lost the book and will be penalized as such).

- 8.4. A rent record whose status is “Returned” or “Book Lost” must be updated. A rent record is finalized and closed when it is given either of those statuses and must not be updated further.

Rental Statuses

1. When a new rental status is registered to the system, it’s given an automatically generated ID by the library’s system.
2. A Book Rental’s Status may only be “On Hold”, “On Time”, “Overdue”, “Returned” or “Book Lost”.

Data Dictionary

Books

Attribute	Datatype	Key	Constraints	Description
Book Barcode	VARCHAR (40)	PK	PRIMARY KEY	Code 39 - type barcode identifying each unique book in the library, including separate copies of the same book.
Title	VARCHAR (200)		NOT NULL	Book title, in full.
Publisher	INT	FK	NOT NULL	The book’s publisher, foreign key referencing the Publishers table.
Publication Date	DATE		NOT NULL	Date of publication for this specific physical book.

Edition	VARCHAR (10)		NOT NULL	The book's edition.
Description	XML			The book's description, synopsis, blurb, etc. describing the book's contents.
Language	INT (250)	FK	NOT NULL	What language this copy of the book is published in.
Alternative Titles	XML			Alternative titles for the book, titles in different languages or different editions.
Genre	INT (50)	FK	NOT NULL	The book's genre, such as fiction, nonfiction, etc.
Availability	INT (4)	FK	NOT NULL	The book's availability, referring to the Availability Statuses table.
Call Number	VARCHAR		NOT NULL	The book's call number/location using the Dewey Decimal System.

Relationships

Table	Type	Description
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Publishers	Many-to-One	Each book has one, and only one, publisher; A publisher has many books.
Languages	Many-to-One	Each book has one, and only one, language; A language has many books written in it.
Genres	Many-to-One	Each book has one, and only one, genre; There are many books in each genre.
Availability Statuses	Many-to-One	Each book is in one, and only one, availability status at any given time; There are many books in the same availability status simultaneously.
Authors	Many-to-Many	Each book can have many, but at least one, author; Each author can have many, but at least one, book.

Rent Records

One-to-Many

Each book can be rented multiple times; Each rent record pertains to one, and only one, book.

Authors

Attribute	Datatype	Key	Constraints	Description
ID	INT	PK	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each author, automatically generated by the system.
First Name	VARCHAR(25)		NOT NULL	Author's first name.
Last Name	VARCHAR(25)		NOT NULL	Author's last name.
Bio	XML			Short biography of the author.
E-mail	VARCHAR(60)		NOT NULL, UNIQUE	Author's contact E-mail.
Phone	VARCHAR(15)		NOT NULL, UNIQUE	Author's contact phone number.
Website	VARCHAR(100)		UNIQUE	Author's website url.

Relationships

Table	Type	Description
Books	Many-to-Many	Each author can have many, but

at least one,
book; Each book
can have many,
but at least one,
author.

Books/Authors

Attribute	Datatype	Key	Constraints	Description
ID	INT	PK	PRIMARY KEY	Unique identifier for each book-to-author relationship, automatically generated by the system.
Book	VARCHAR (40)	FK	NOT NULL	The book's barcode, foreign key referencing the Books table.
Author	INT	FK	NOT NULL	The author's ID, foreign key referencing the Authors table.
Relationships				
Books/Authors is the junction table related to the Many-to-Many relationship between Books and Authors				

Publishers

Attribute	Datatype	Key	Constraints	Description
ID	INT	PK	PRIMARY KEY	Unique identifier for each publisher, automatically generated by the system.

Name	VARCHAR (50)	NOT NULL	The publisher's name, in full.
Website	VARCHAR(100)	NOT NULL, UNIQUE	The publisher's business website.
E-Mail	VARCHAR(60)	NOT NULL, UNIQUE	Publisher's contact E-mail.
Phone	VARCHAR(15)	NOT NULL, UNIQUE	Publisher's contact phone number.
Address	VARCHAR(150)	NOT NULL, UNIQUE	Publisher's physical address.
Date of Founding	DATE	NOT NULL	Date when the publisher was founded.

Relationships

Table	Type	Description
Books	One-to-Many	Each book has one, and only one, publisher; Each publisher can have many books.

Languages

Attribute	Datatype	Key	Constraints	Description
ID	INT (250)	PK	PRIMARY KEY	Unique identifier for each language, automatically generated by the system.
Language	VARCHAR (25)		NOT NULL, UNIQUE	Language, for example English, Irish, French, etc.

Relationships		
Table	Type	Description
Books	One-to-Many	Many books can be published in a language; Each book is published in one, and only one, language.

Book Genres				
Attribute	Datatype	Key	Constraints	Description
ID	INT (50)	PK	PRIMARY KEY	Unique identifier for each genre, automatically generated by the system.
Genres	VARCHAR (15)		NOT NULL, UNIQUE	Boom genre, for example, fiction, nonfiction, biography, anatomy textbook, etc.
Relationships				
Table	Type		Description	
Books	One-to-Many		There can be many books in one genre; Each book has one, and only one, genre.	

Availability Statuses				
Attribute	Datatype	Key	Constraints	Description
ID	INT (4)	PK	PRIMARY KEY	Unique identifier for each availability

Status	VARCHAR(25)	NOT NULL, UNIQUE	status, automatically generated by the system. The book's availability status, "Available", "Rented", "On Hold" or "Unavailable for Renting".
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Relationships

Table	Type	Description
Books	One-to-Many	There are many books with the same availability status simultaneously; A book must be in one, and only one, availability status at a time.

Members

Attribute	Datatype	Key	Constraints	Description
ID	INT	PK	PRIMARY KEY	Unique identifier for each member, automatically generated by the system.
First Name	VARCHAR(25)		NOT NULL	Member's first name

Last Name	VARCHAR(25)		NOT NULL	Member's last name
E-Mail	VARCHAR(60)		NOT NULL, UNIQUE	Member's contact E-Mail.
Phone	VARCHAR(15)		NOT NULL, UNIQUE	Member's contact phone number.
Membership Type	INT (8)	FK	NOT NULL	Type of membership, Foreign Key to the Membership Types table.
Membership Active	BOOL		NOT NULL	Whether the member is active, that is, allowed to borrow books or not.
Date of Joining	DATE		NOT NULL	Date in which the member was registered in the system.

Relationships

Tables	Type	Description
Membership Type	Many-to-One	Each member has only one membership type at a time; Multiple members might have the same membership type simultaneously.
Rent Records	One-to-Many	The same

member may rent multiple books; A record pertains only to one instance of one specific member renting a book.

Membership Types

Attribute	Datatype	Key	Constraints	Description
ID	INT (8)	PK	PRIMARY KEY	Unique identifier for each membership type, automatically generated by the system.
Membership Type	VARCHAR (20)		NOT NULL, UNIQUE	The membership type, “Child”, “Young Adult”, “Adult”, “Student”, “Professor”, “Researcher” or “Visitor/Tourist”

Relationships

Table	Type	Description
Members	One-to-Many	Multiple members might have the same membership type simultaneously;

Each member has only one membership type at a time.

Staff

Attribute	Datatype	Key	Constraints	Description
Employee ID Number	VARCHAR (9)	PK	PRIMARY KEY	Unique identifier for each library employee, not created by the database but by the system in which the employees are enrolled through.
First Name	VARCHAR (25)		NOT NULL	Employee/staff member's first name.
Last Name	VARCHAR (25)		NOT NULL	Employee/staff member's last name.
Hiring Date	DATE		NOT NULL	Date in which the employee/staff member was hired.
Date of Birth	DATE		NOT NULL	Employee/staff member's date of birth.
E-Mail	VARCHAR (60)		NOT NULL, UNIQUE	Employee/staff member's contact E-mail.

Phone	VARCHAR (15)		NOT NULL, UNIQUE	Employee/staff member's contact phone number.
Credentials	VARCHAR (15)	FK	NOT NULL, UNIQUE	Employee/staff member's system login credentials, foreign key referencing the Credentials table.
Role	VARCHAR (25)		NOT NULL	Employee/staff member's role in the library team.

Relationships

Tables	Type	Description
Credentials	One-to-One	Each staff member has one, and only one, set of login credentials; Each set of login credentials pertains to one, and only one, staff member.
Rent Records	One-to-Many	Each staff member may authorize multiple rentals; Each rent record is authorized by one, and only one, staff member.

Credentials

Attribute	Datatype	Key	Constraints	Description
Username	VARCHAR (15)	PK	PRIMARY KEY	Unique login username used to access the library system.
Password	CHAR (256)		NOT NULL	Sha-256 hash calculation of the password associated with that login.
Permission Level	INT (15)		NOT NULL	That login's permission level within the system, foreign key referencing the User Permission Levels table.
Credentials Active	BOOL		NOT NULL	Whether the credentials are active in the system or not.

Relationships

Table	Type	Description
Staff	One-to-One	Each set of login credentials pertains to one, and only one, staff member; Each staff member has one, and only one, set of login credentials.

User Permission Levels	Many-to-One	A set of credentials must be associated with one, and only one, level of permission at any time; A level of permission can be associated with multiple credentials simultaneously.
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User Permission Levels

Attribute	Datatype	Key	Constraints	Description
ID	INT (15)	PK	PRIMARY KEY	Unique identifier for each user permission level, automatically generated by the system.
Permission Level	VARCHAR (25)		NOT NULL	Permission level within the system, each permission level indicates what users in that level are allowed to do, ex. rent and return books, register a new member, change publisher information, change other users' login

permission levels.

Relationships

Table	Type	Description
Credentials	One-to-Many	A level of permission can be associated with multiple credentials simultaneously; A set of credentials must be associated with one, and only one, level of permission at any time.

Rent Records

Attribute	Datatype	Key	Constraints	Description
ID	INT	PK	PRIMARY KEY	Unique identifier for each rent record, automatically generated by the system.
Rented By	INT	FK	NOT NULL	Member that rented the book, foreign key referencing the Members table.
Rented Book	VARCHAR (40)	FK	NOT NULL	Barcode of the book that was rented, foreign key referencing the Books table.
Date of Rental	DATETIME		NOT NULL	Date in which this rental record was made.

Due Date	DATETIME		NOT NULL	Date by which the book must be returned to avoid fines.
Authorized By	VARCHAR (9)	FK	NOT NULL	Employee/staff member that authorized the book rental, foreign key referencing the Staff table.
Status	INT (5)	FK	NOT NULL	Status of the rental, foreign key referencing the Rental Statuses table.
Relationships				
Table		Type		Description
Members		Many-to-One		A record pertains only to one instance of one specific member renting a book; The same member may rent multiple books.
Books		Many-to-One		Each rent record pertains to one, and only one, book; Each book can be rented multiple times.
Staff		Many-to-One		Each rent record is authorized by one, and only one, staff member; Each staff member may authorize multiple rentals.
Rental Statuses		Many-to-One		Each rent record is at one, and only one, status at one time; Many rent records may have the same status at the same time.

Rental Statuses

Attribute	Datatype	Key	Constraints	Description
ID	INT(5)	PK	PRIMARY KEY	Unique identifier for each rental status, automatically generated by the system.
Status	VARCHAR (10)		NOT NULL, UNIQUE	The rental status, "On Hold", "On Time", "Overdue", "Returned" or "Book Lost".
Fine	VARCHAR (20)			Fine associated with the rental status, by total owed or amount owed per day of lateness.

Relationships

Table	Type	Description
Rent Records	One-to-Many	Each rent record is at one, and only one, status at one time; Many rent records may have the same status at the same time.

Conclusion

In summary, this project effectively shows how relational database design ideas may be applied to an actual organizational setting. A structured and normalized data model that represents the organization's operational requirements was produced by creating an entity-relationship diagram for a library management system. The relationships between core entities such as books, publishers, authors, members, etc., are precisely defined by the ERD.

The design process highlighted how important careful planning is to design databases. The overall integrity and effectiveness of the data model were significantly impacted by choices made on entity identification, attribute assignment, and relationship cardinality. The resulting schema is scalable and maintainable, able to meet the expanding requirements of a library by following normalization principles and removing unnecessary data.

Overall, this project offered significant practical experience in data modelling while strengthening the theoretical concepts learned. The skills gained in analyzing organizational data requirements and converting them into a well-structured ERD are relevant to professional database development. Future developments will focus on integrating the ERD schema into a relational database management system such as MySQL.

Individual contributions

Alexandre Peres Oliveira da Silva

Work done for this project was done, for most of it, collectively. The group got together for meetings multiple times a week, in which each member contributed equally to the discussion and design of the database in question; decisions were made after consulting every group member and collaborative software (Google Docs, LucidChart) was used, so that group members could add to and edit the project concurrently.

Initially, after each member of the group researched possible entities and companies that could be used as the project's focus, a meeting was held, and each group member shared their ideas for what sort of database could be used for the project. Other than a library management system, which was, in the end, chosen by popular vote, other possibilities for database projects included a gym membership management system, a food delivery app, an E-commerce website and a music app. For each of these ideas, group members suggested what kind of tables and scope would be applicable, not yet with much detail, in order to help inform each group member's choice. After that meeting, each group member thought on which idea they favored, ranking them 1 to 5, with the idea they favored the most getting 5 points, and the

idea they favored the least, 1 point. The idea of a library management system won, with 13 points total across all three members' rankings.

In the following meeting, the idea for a library's system was further polished, the tables were further elaborated and polished through discussion between group members. Scope and limitations of the systems were discussed, for example, it was decided that entities containing an email and a phone number field should only have one of each, which should be kept up to date by each entity, instead of having separate "phone numbers" and "E-mails" tables serving multiple entities, considering that keeping multiple forms of contact would not be particularly helpful. One particular notable suggestion given by group member Alexandre was the addition of an "Alternative titles" variable attribute to the "Books" table, storing the books' titles in other languages, if it has any, which would be helpful to fulfill the project's requirement for a hybrid database system.

Another relevant aspect of the work done by Alexandre was in regards to the understanding of library classification systems, and what would be the best information to use as a book's primary key identifier. After the problem with using the ISBN, due to it being the same across multiple copies of the same book, was brought up by the group, it was the responsibility of the member in question to research an alternative. The decision to use a barcode, and the specific type of barcode, was made as a result of the research done, as well as the decision to keep a call number based on the Dewey Decimal System stored as a non-key attribute.

The creation of the report itself was done through a more individual approach; since the database and business rules had already been designed and created collectively, writing the report would be more a matter of putting those agreed upon ideas in text format, which could be more efficiently done by dividing the work. The group member in question, Alexandre, was mainly responsible for writing the Organisation Overview section and the Business Rules section.

Finally, Alexandre was also responsible for formatting and reviewing language and grammar on the report, and polishing the text in other sections of the report.

Okeke Munachukwudinamma Praise

Just like my group partners said, most of the project work was completed collaboratively, decisions were made collectively after consulting all members, and we met up a couple of times a week, using discord to discuss and work on the project. The presented project was agreed

upon through a voting system after we looked through multiple project ideas such as a music app, a food delivery app, gym membership management system and so on.

After agreeing to work on a library for our project, I contributed to the development of the business rules and data dictionary sections of the report. This involved defining the constraints, relationships, and attribute requirements for the various entities within the library database. I also joined in ensuring consistency and integrity across the data model throughout the design process.

I also assisted in the design and development of the Entity-Relationship Diagram. It involved identifying the core entities of the library management system and defining their attributes. This includes entities such as Books, Members, Staff, Authors and Publishers, and they all require careful analysis in terms of what information the database would need to store. For example, the Books entity was assigned attributes such as a barcode, title, publication date, edition, language, genre and so on, while the Members entity included attributes such as member ID, contact information, and membership type. I also joined in creating relationships and cardinalities between the various entities, such as one-to-one relationships, many-to-one relationships, for example Rent Records and Members, which shows that one member can rent multiple books or the relationship between Books and Authors, which required a table to account for the fact that a book may have multiple authors and an author may have written multiple books, etc. The primary design tool used which allowed the group to collaboratively build and refine the diagram is LucidChart.

I contributed to the writing of the introduction and conclusion sections of the report. The introduction involved contextualizing the purpose of the assessment, outlining the chosen organisation, and establishing the objectives of the database design. I also drafted the conclusion, which required synthesizing the outcomes of the project, reflecting the design process, and highlighting the practical and theoretical value of the work undertaken.

Spencer Ibhafidon John

During the course of this continuous assessment project, I took part in both the research and design stages of the Library database system. I joined the group members in brainstorming for business ideas and finalising the organisation for our project, which was done by polling. I also helped in analysing how the organisation functions and what data we might need to work with, and also which business transactions will be supported by the database design.

An important part of my teamwork was helping to create the Entity Relationship Diagram (ERD). I shared some ideas on how many entities the design might need, and also their

respective attributes, and how tables should connect with primary and foreign keys. I also made sure our ERD design followed database normalisation rules and met the business rules we all identified.

Aside from helping design the ERD diagram used in this project, I participated in creating the diagram using Lucidchart. I made sure the Diagram corresponds with the database structure and the assessment's requirements. I shared ideas about the entities and their attributes, and how tables should connect with primary and foreign keys.

I also researched the most suitable means to report this project more professionally, such as a research template and also stuck to the formatting styles from the assessment requirements. I also reviewed a couple of examples of professional database documentation and informed the team about the font style, spacing, layout, and whole document, making sure it's consistent with the requirements.

Additionally, I help in creating the referencing and developing the referencing guide and proofreading the report for spelling mistakes. I checked sections written by my teammates to make the report cleaner, easier to read, and more presentable, which made it possible for the report to be very structured. I helped in setting up the medium of communication for the team; we used email and sometimes Discord calls and planning.

Furthermore, I helped with creating some part of the data dictionary, I worked on defining table attributes, data types and constraints for several database entities which assisted in maintaining the consistent of the ERD and database documentations. My involvement assisted in improving the database design and made the report more polished and refined.

References

Saxena, M. (2023) *15 Interesting Database Project Ideas for Students*. Medium, 9 October.

Available at:

<https://medium.com/@mohinisaxena/15-interesting-database-project-ideas-for-students-7842941a8930> (Accessed: 11 May 2026).

Dinantio, P. (2024) *Creating a Relational Database and Constructing an Entity Relationship Diagram for E-Libraries*. Medium, 1 May. Available at:

<https://medium.com/@dinantio/creating-a-relational-database-and-constructing-an-entity-relationship-diagram-for-e-libraries-9a6bb1375c40> (Accessed: 14 May 2026).

BM (no date) *Types of business rules*. IBM Documentation. Available at:
<https://www.ibm.com/docs/en/iis/11.7.0?topic=types-business-rules> (Accessed: 22 May 2026).

Librarianship Studies and Information Technology (2015) *Library classification*. Available at:
<https://www.librarianshipstudies.com/2015/08/library-classification.html> (Accessed: 24 May 2026).

University of Limerick Library (2026) *What is a call number?* Available at:
<https://ul-ie.libanswers.com/faq/277449> (Accessed: 24 May 2026).

Barcode1 Ireland (no date) *Types of barcodes*. Available at:
<https://barcode1.ie/types-of-barcodes/> (Accessed: 24 May 2026)

Yale University Library (no date) *Call numbers for books*. Available at:
<https://web.library.yale.edu/cataloging/music/call-numbers-for-books> (Accessed: 24 May 2026).

Yale University Library (no date) *Call Numbers for Books*. Available at:
<https://web.library.yale.edu/cataloging/music/call-numbers-for-books> (Accessed: 24 May 2026).

W3Schools (no date) *SQL Data Types for MySQL, SQL Server, and MS Access*. Available at:
https://www.w3schools.com/sql/sql_datatypes.asp (Accessed: 24 May 2026).